2. Theory / Conceptual question

As discribed in the question, this should be a Geometric Distribution.

2a) the first success occurs at the kth trial, so $n=k$ with $k-1$ failure in strike and 1 success.

$$p(x=k) = (1-p)^{k-1} \cdot p$$

$$\therefore L(p) = \prod_{i=1}^{n} f(x|p) = \prod_{i=1}^{n} (1-p)^{x_i-1} \cdot p \quad x_i \in [1, k]$$

$$\therefore \ell L(p) = \ln \prod_{i=1}^{n} (1-p)^{x_i-1} \cdot p$$

$$= \sum_{i=1}^{n} \ln p + \sum_{i=1}^{n} (x_i-1) \cdot \ln(1-p)$$

in order to gain maximum of $\ell(p)$, let

$$\frac{\partial}{\partial p} \ell(p) = 0$$

$$\frac{\partial}{\partial p} \left\{ \sum_{i=1}^{n} \ln p + \sum_{i=1}^{n} (x_i-1) \ln(1-p) \right\} = 0$$

$$\therefore \frac{n}{p} - \frac{\left(\sum_{i=1}^{n} x_i - n\right)}{1-p} = 0$$

$$n(1-p) - p\sum_{i=1}^{n} x_i + np = 0$$

$$\therefore p = \frac{n}{\sum_{i=1}^{n} x_i} \Rightarrow \hat{p} = \frac{1}{\frac{\sum_{i=1}^{n} x_i}{n}} = \frac{1}{k} \quad (n=k)$$

2b) Since 2a) expriment is independent, so we can say that the serial experient conducted in 2b) is also independent

$$f(x;p) = (1-p)^{x_{ij}-1} p^1 \qquad L(n) = (1-p)^{\sum_{i=1}^{n}\sum_{j=1}^{k} (x_{ij}-1)} \cdot p^n$$

$$\therefore \ell L(p) = \sum_{i=1}^{n} \ln p + \sum_{i=1}^{n}\sum_{j=1}^{k} (x_{ij}-1) \cdot \ln(1-p)$$

$$\therefore \frac{\partial}{\partial p} \left\{ \sum_{i=1}^{n} \ln p + \sum_{i=1}^{n}\sum_{j=1}^{k} (x_{ij}-1) \ln(1-p) \right\} = 0$$

$$\Rightarrow \hat{p} = \frac{1}{\frac{\sum_{j=1}^{k}\sum_{i=1}^{n} x_{ij}}{n}} = \frac{n}{k_1 + k_2 \cdots + k_n}$$

3. Theory / Conceptual question

3a) The pdf of each observation should has the following form $f(x|\alpha) = \begin{cases} \frac{1}{\alpha} & \text{for } 0 \le x \le \alpha \\ 0 & \text{otherwise} \end{cases}$

so the likelyhood function should be

$$L(\alpha) \begin{cases} \frac{1}{\alpha^m} & 0 \le x_i \le 0 \quad i \in [1, m] \\ 0 & \text{otherwith} \end{cases}$$

we all know that $\alpha \ge x_i$, since $\alpha > 0$, we should find $\hat{\alpha} = \max(X_1, \cdots X_m)$, to achieve max $L(\alpha)$ since it's a decreasing function and $\alpha \ge x_i$

3b) the current pdf is $f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta - \alpha}, & x \in [\alpha, \beta] \\ 0, & \text{otherwise} \end{cases}$

$\therefore L(\alpha|\beta) = \frac{1}{(\beta-\alpha)^m}$

to achieve maxmize $L(\alpha|\beta)$, we should make $(\beta - \alpha)$ as small as possible, which means we should pick $\min\{x_1, \cdots, x_m\}$ for β and $\max\{x_1, \cdots, x_m\}$

since $x_i \ge \alpha$, $x_i \le b$